Lesson 12 Teacher's Key: Hearing It Happen

GameMaker - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

No new GameMaker mechanism this week - audio_play_sound() was already fully established by Lesson 3, so this lesson is purely about recognizing which existing branches (from Lessons 4, 5, 10, 11) are the correct trigger points for two brand-new sounds.

Level 1 - Full Worked Solution

- sfx_hit moves to directly follow the score/enemies_remaining updates in obj_bullet's Collision event - matching the same restructured location every other engine's hit sound now lives in.
- sfx_wave and sfx_lose are the two genuinely new sounds, placed exactly where Lesson 11's nested if and Lesson 8's lives-check already live.
- audio_play_sound()'s three arguments (sound, priority, loop) stay identical to Lesson 3's original call for every one of the five sounds this week - no new parameter usage needed.

Why This Structure

- Because every trigger point (a collision, a wave transition, an escape) is already event-driven rather than polled, none of today's five sounds need any kind of replay-guard - they each fire from code that only runs once per actual occurrence.
- Keeping every sound call physically adjacent to the game-state change it represents (score/enemies_remaining update, then immediately the matching sound) keeps this lesson's additions easy to audit later - a direct continuation of the Structure principle from Lesson 7's original bugfix.

Level 2 / Level 3 Talking Points

- Ask students to find all five audio_play_sound() calls across obj_bullet and obj_enemy and name, for each, exactly which player action or game event causes it to run.
- For advanced students: ask what would need to change to give each enemy type (if the project ever added more than one) its own distinct hit sound - not required this week, a natural preview of expanding the Sound resource set later in the year.

Common Student Mistakes

- Placing the sfx_wave call outside the else branch, causing it to also fire on the true win.

- Forgetting the obj_spawner. prefix somewhere in the revised blocks, since this code still runs from obj_bullet and obj_enemy.
- Creating a Sound resource but never actually assigning an imported .wav file to it, leading to a silent audio_play_sound() call with no error.